# ABRAAR KHAN

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## PROFILE

- Software engineer shipping production RAG, agentic, and document intelligence systems in Python and TypeScript
- Built secure FastAPI backends for sensitive data with PostgreSQL Row-Level Security and PHI masking
- Codes daily with Claude Code and Cursor while owning review, tests, and release quality for each change

## EXPERIENCE

**Data Scientist** Jun 2026 to Present

*CDF, Sparks, NV*

- Cut Medicaid chart review 25%, saving 15 hours weekly, by building a clinical RAG summarization engine
- Raised retrieval accuracy from 72% to 88% across 50K+ forms by engineering hybrid keyword and FAISS search
- Reduced response latency 2.4s by chunking long assessment forms into clinical sections within context limits
- Lifted answer reliability to 93% and halved unsupported claims via physician-graded LLM evaluation pipelines
- Protected 60K+ patient lookups from leakage by adding guardrails that mask PHI and verify medical terms

**ML and Data Engineer Intern** Aug 2025 to May 2026

*Aventusoft (early-stage startup, UF IPPD), Boca Raton, FL*

- Enabled real-time cardiac diagnosis on wearables by deploying a streaming ECG inference service in PyTorch
- Improved arrhythmia F1 14% and AUC 11% by synthesizing rare-class waveforms to balance training sets
- Kept every model update audit-ready by enforcing IEC 62304 compliant CI/CD release gates
- Increased dashboard adoption among cardiologists by redesigning the Chart.js frontend in JavaScript

**AI and Analytics Engineer Intern** Jan 2023 to Aug 2023

*Plumb5 (startup), Bengaluru, India*

- Increased result precision 12% for multi-region enterprise clients by shipping FAISS multi-vector reranking
- Eliminated per-request connection overhead on a latency-bound Postgres backend via Python connection pooling
- Shortened external developer onboarding by authoring REST API contracts and JSON Schema specifications
- Cleaned 10+ ingestion streams before downstream use by automating outlier detection and validation checks

## PROJECTS

**Multi-Agent Data Assistant |** github.com/abraar64mak/mcp-data-assistant

*Python, LangGraph, MCP, FastAPI, PostgreSQL, OpenTelemetry*

- Achieved 75% execution accuracy and 100% adversarial refusal by orchestrating NL-to-SQL agents in LangGraph
- Blocked all writes, injections, and unauthorized reads across 65 attack tests with a sqlglot AST firewall
- Isolated tenant data through role-based table access, query cost budgets, and PostgreSQL row policies
- Made each answer auditable by returning executed SQL, per-agent timings, and OpenTelemetry trace IDs

**Transformer Document Intelligence Platform |** github.com/abraar64mak/document-intelligence-platform

*Python, Hugging Face Transformers, PySpark, FastAPI*

- Reached 0.916 strict entity F1 on 3,453 CoNLL-2003 test samples by fine-tuning RoBERTa behind a CI gate
- Scaled distributed batch extraction by fanning NER inference across PySpark workers loading weights once
- Kept jobs running through corrupt inputs by bisecting crashed batches and quarantining bad files
- Served up to 32 documents per request via an endpoint returning entities, confidence scores, and offsets

## SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, Bash
**AI and LLM Systems:** PyTorch, Hugging Face Transformers, LangGraph, MCP, RAG, FAISS, Agents, LLM Evaluation, Ragas
**Backend and Data:** FastAPI, REST APIs, PostgreSQL, PySpark, JSON Schema, Vector Databases, Distributed Systems
**Cloud and DevOps:** AWS, Azure, Docker, GitHub Actions, CI/CD, RBAC/RLS
**AI-Assisted Development:** Claude Code, Cursor

## EDUCATION

**University of Florida** Aug 2024 to May 2026

*Master of Science in Artificial Intelligence Systems, Gainesville, FL*

## AWARDS

2x Best Paper Award for peer-reviewed healthcare AI research: ICCCNet 2023 (Springer LNNS), IConIC 2K23 (AIP)